

KINGSTON 1 NW, RI, USA

WMO: 725110

Lat: 41.4911N

Lon: 71.5413W

Elev: 115

StdP: 14.63

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 54796

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 5.4	(c) 10.4	(d) -8.1	(e) 3.6	(f) 8.6	(g) -3.0	(h) 4.7	(i) 13.9	(j) 20.4	(k) 31.5	(l) 17.7	(m) 32.6	(n) 2.5	(o) N/A	(p) 0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 16.7	(c) 86.3	(d) 73.4	(e) 83.3	(f) 71.6	(g) 80.8	(h) 70.3	(i) 75.9	(j) 82.7	(k) 74.5	(l) 80.3	(m) 73.2	(n) 78.3	(o) 5.4	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.8	126.5	79.5	72.8	122.0	78.0	71.4	116.6	76.4	39.4	83.1	38.0	80.2	36.9	78.3	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 15.4	(b) 12.9	(c) 11.0	(e) -2.5	(f) 90.9	(g) 6.2	(h) 2.9	(i) -6.9	(j) 93.0	(k) -10.6	(l) 94.7	(m) -14.1	(n) 96.3	(o) -18.6	(p) 98.5		
			DB	-4.2	79.1	5.8	1.8	-8.4	80.4	-11.8	81.5	-15.0	82.5	-19.2	83.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.0	28.7	30.1	36.8	46.9	56.5	64.7	71.3	69.9	63.4	52.7	42.9	34.5	(6)
(7)		DBStd	16.56	10.11	8.93	8.47	7.09	6.81	5.96	4.85	5.11	6.18	7.78	8.24	8.89	(7)
(8)		HDD50	2583	662	558	415	142	15	0	0	0	0	61	244	486	(8)
(9)		HDD65	5996	1126	977	874	543	277	77	7	14	103	388	664	946	(9)
(10)		CDD50	2568	1	1	6	49	216	441	659	615	401	144	30	5	(10)
(11)		CDD65	508	0	0	0	1	14	68	201	164	54	6	0	0	(11)
(12)		CDH74	3313	0	0	0	27	128	479	1337	1084	237	21	0	0	(12)
(13)		CDH80	709	0	0	0	6	27	104	326	217	27	2	0	0	(13)
(14)	Wind	WSAvg	4.5	4.5	4.6	5.3	5.4	5.0	4.3	3.9	3.5	4.0	4.5	4.3	4.3	(14)
(15)	Precipitation	PrecAvg	51.2	4.3	3.8	4.9	4.6	4.2	3.7	3.1	4.1	4.3	4.1	5.0	4.8	(15)
(16)		PrecMax	70.2	12.4	8.3	12.3	13.0	9.6	14.3	7.4	12.7	10.9	14.9	13.0	11.3	(16)
(17)		PrecMin	32.4	0.9	0.9	0.7	1.2	0.7	0.0	0.7	0.6	0.9	0.6	0.7	0.9	(17)
(18)		PrecStd	7.9	2.3	1.7	2.4	2.4	2.2	2.9	1.9	2.7	2.1	2.4	2.5	2.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.2	56.6	66.0	77.7	83.6	87.9	90.6	88.7	83.8	77.3	66.5	60.9	(19)
(20)			MCWB	54.5	51.4	54.5	62.0	68.4	73.1	76.6	75.7	73.7	65.5	60.0	57.6	(20)
(21)		2%	DB	52.9	52.1	58.0	68.5	77.2	82.8	86.6	85.0	79.2	71.3	63.1	56.9	(21)
(22)			MCWB	50.0	47.9	50.4	56.4	65.4	69.8	73.9	73.0	69.3	66.2	58.8	54.7	(22)
(23)		5%	DB	48.4	48.4	53.6	63.2	73.0	79.3	83.6	82.2	76.7	68.5	60.2	53.7	(23)
(24)			MCWB	45.6	44.3	48.1	53.9	62.8	67.6	72.4	71.2	68.8	63.6	55.5	51.4	(24)
(25)		10%	DB	43.9	44.2	50.2	59.1	68.6	76.2	80.9	79.9	74.2	65.9	57.7	49.9	(25)
(26)			MCWB	40.3	39.8	44.9	51.9	59.7	65.3	71.1	70.4	68.0	61.4	52.9	46.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.9	54.6	59.1	62.9	71.4	75.1	78.9	77.9	75.5	70.2	62.7	59.3	(27)
(28)			MCDWB	56.9	56.6	65.6	70.9	81.6	83.9	88.4	85.6	81.7	73.3	65.1	60.1	(28)
(29)		2%	WB	51.1	50.4	52.3	59.0	68.4	72.4	76.4	75.5	73.4	67.7	60.3	56.0	(29)
(30)			MCDWB	52.4	52.6	55.8	62.0	75.6	80.1	84.4	81.0	78.1	70.7	62.2	57.2	(30)
(31)		5%	WB	45.9	46.2	49.6	56.3	65.5	70.3	74.8	74.3	71.5	65.3	57.0	52.9	(31)
(32)			MCDWB	48.3	49.2	53.7	60.3	71.0	76.3	81.3	78.9	74.9	68.4	59.4	54.5	(32)
(33)		10%	WB	41.2	41.6	45.3	53.5	63.0	68.5	73.5	72.9	70.1	62.9	53.0	48.7	(33)
(34)			MCDWB	43.6	45.1	49.1	57.3	68.0	73.6	79.1	77.0	73.2	66.2	56.5	51.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.1	17.3	17.2	18.3	18.0	17.2	16.7	17.4	18.2	18.3	17.9	15.9	(35)
(36)			MCDBR	20.8	21.1	21.7	24.7	23.6	21.7	19.3	19.0	19.3	19.1	19.3	18.6	(36)
(37)			MCWBR	20.2	16.8	14.3	12.6	12.3	10.9	8.9	9.6	11.3	13.1	17.0	17.7	(37)
(38)		5% WB	MCDWB	22.1	21.6	19.2	17.1	19.4	18.4	16.1	15.3	14.7	15.8	18.2	18.1	(38)
(39)			MCWBR	21.4	19.3	15.2	12.7	11.5	10.3	8.3	9.1	9.7	12.9	17.7	18.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.305	0.313	0.336	0.374	0.418	0.461	0.485	0.445	0.376	0.343	0.315	0.303	(40)	
(41)		taud	2.500	2.477	2.454	2.381	2.288	2.208	2.149	2.254	2.421	2.520	2.557	2.556	(41)	
(42)		Ebn at Noon	271	285	290	285	273	260	252	259	272	271	265	263	(42)	
(43)		Edh at Noon	23	28	32	37	41	45	47	41	33	26	22	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	599	877	1198	1547	1710	1876	1922	1725	1407	952	651	502	(44)	
(45)		RadStd	47	86	86	166	164	189	101	91	104	87	47	49	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.10	N/A	N/A	+1.44	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+173	+85

Nomenclature: See separate page